



Prepare 50% Faster

Percentage Part 4

Assignment with Solutions

1. When income of a man decreases by Rs. 4000 then tax rate increases from 10% to 15%. While in both the situation 15% income is tax free. Find his initial income if he paid equal tax in both cases.

1) 18500

2) 16500

3) 17000

4) 12000

5) 19000

Ans: 4) 12000

2. A salesman is allowed 24% commission on the total sales by him and a bonus of 1% on the sales over Rs. 25000. If the total earnings of a salesman is Rs. 11475. Find the total sales.

1) 46,900

2) 48,400

3) 44,800

4) 44,560

5) None of these

Ans: 1) 46,900

3. A salesman is allowed 16% commission on the total sales by him and a bonus of 5% on the sales over Rs. 20000. If the total earnings of a salesman is Rs. 10865. Find the total sales.

1) 85000

2) 75400

3) 56500

4) 45210

5) None of these

Ans: 3) 56500

4. A company allowed 9% commission on the total sales made by the salesman and a bonus of 1% on the sales over Rs. 40000. If the salesman deposited Rs. 108400 in the company after deducting his earnings from the total sales made by him. Find the total sales?

1) Rs. 120000

2) Rs. 390000

3) Rs. 180000

4) Rs. 150000

5) None of these

Ans: 1) Rs. 120000

5. A company allowed 16% commission on the total sales made by the salesman and a bonus of 1% on the sales over Rs. 25000. If the salesman deposited Rs. 66650 in the company after deducting his earnings from the total sales made by him. Find the total sales?

1) Rs. 60000

2) Rs. 80000

3) Rs. 92000

4) Rs. 108000

5) None of these

Ans: 2) Rs. 80000

6. A Company allow 8% commission on the total sales to its salesman but if the salesman is appointed on a fix salary of Rs. 10000 and 6% commission on the sales more than 30000. If in second condition the salesman receives 2400 more than the first condition. Find the total sales?

1) 255000

2) 335000

3) 265000

4) 290000

5) 328000

Ans: 4) 290000

7.A Salesman is allowed 18% commission on the total sales made by him and a bonus of 1% on the sales over Rs. 40000. If the total earning of a salesman is Rs. 11000. Find the total sales.

1) 75000

2) 60000

3) 80000

4) 120000

5) None of these

Ans: 2) 60000

8.If a student scores 30% marks then he is failed by 50 marks, but when he scores 50% marks then he is passed by 30 marks. Find the passing %?

1) 42.5%

2) $33\frac{1}{3}\%$

3) 37.5%

4) 45%

5) None of these

Ans: 1) 42.5%

9.A student has to secure 40% marks to pass an examination. He gets 150 marks and fails by 30 marks. What are the maximum marks?

1) 400

- 2) 500
- 3) 475
- 4) 450
- 5) None of these

Ans: 4) 450

10. In an election between two candidates A & B 10% voters didn't participate in voting. 700 casted votes declared invalid and the winner got 70% of the total valid votes and won by 800 votes. Then find the total number of voters in the voting list.

- 1) 4000
- 2) 4200
- 3) 3200
- 4) 3000
- 5) None of these

Ans: 4) 3000

11. In an election there are a total of 4,000 registered voters out of which $x\%$ of voters did not cast their votes and 180 votes are declared invalid. The winner got 740 votes more than the losing candidate. The number of valid votes received by the winner are 42% of the total registered voters. How many registered voters did not cast their vote?

- 1) 1200
- 2) 1500
- 3) 350
- 4) 1250
- 5) None of these

Ans: 1) 1200

12. There were two candidates in an election. 15% of the total voters did not cast their votes whereas 10% of the casting votes were declared invalid. If the winning candidate

received 60% of the total votes and won by 3,480 votes then find out the total number of Voters?

- 1) 5,000
- 2) 8,000
- 3) 6,000
- 4) 4,000
- 5) None of these

Ans: 2) 8,000

13. In an election two candidates participating 20% voters did not cast their voters, out of which 1200 votes declared invalid and the winner get 60% votes of valid votes and he wins by 3760 votes. Find the no. of voting list.

- 1) 25000
- 2) 20000
- 3) 22000
- 4) 18000
- 5) 24000

Ans: 1) 25000

14. A student multiplied a number by $\frac{4}{5}$ instead of $\frac{5}{4}$. What is the percentage error in the calculation?

- 1) 27%
- 2) 64%
- 3) 36%
- 4) 24%
- 5) None of these

Ans: 3) 36%

15. The ratio of the expenditure of a person on food rent and other things are 5: 2: 3 respectively. There is an increase in his expenditure of 10%, 25% & 30% respectively. So how much percentage increased in his expenditure?

- 1) 18%
- 2) 19%
- 3) 17%
- 4) 20%
- 5) 24%

Ans: 2) 19%

Solution

Solution:1 - Let the initial income is x after increase of 4000 it will be $x-4000$

$$x \times 85/100 \times 10/100 = (x-4000) \times 85/100 \times 15/100$$

On solving we get $x \Rightarrow 12000$ ans.

Solution:2-

Let the total sales be x

According to the question

$$x \times (24/100) + (x - 25000) \times (1/100) = 11,475$$

$$24x/100 + (x - 25000)/100 = 11475$$

$$24x + x - 25000 = 11,47,500$$

$$25x = 11,72,500$$

$$x = 46,900$$

Solution:3- Let the total sales be x

According to the question

$$x \times 16/100 + (x - 20000) \times 5/100 = 10865$$

$$(16x + 5x - 100000)/100 = 10865$$

$$21x - 100000 = 1086500$$

$$21x = 1186500$$

$$x = 56500$$

Solution:4-

Let the total sales be x

$$\text{Earning of the salesman} = x \times \frac{9}{100} + (x - 40000) \times \frac{1}{100} = \frac{9x}{100} + \frac{x}{100} - 400 = \frac{10x}{100} - 400$$

Now, According to the question,

$$x - \left(\frac{10x}{100} - 400 \right) = 108400$$

$$x - \frac{10x}{100} + 400 = 108400$$

$$\frac{90x}{100} = 108000$$

$$x = 120000$$

Solution:5- Let the total sales be Rs. x

$$\text{Earning} = x \times 16\% + (x - 25000) \times 1/100$$

$$= 16x/100 + x/100 - 250 = 17x/100 - 250$$

Now, According to the question

$$\text{Total Sales} - \text{Earning} = 66650$$

$$x - (17x/100 - 250) = 66650$$

$$x - 17x/100 + 250 = 66650$$

$$83x/100 = 66400$$

$$x = 80000$$

Solution:6- Let the total sales be x

$$\text{1st Condition Salesman gets} = x \times 8\%$$

$$\text{2nd Condition Salesman gets} = 10000 + (x - 30000) \times 6\%$$

The difference of 2nd and 1st condition is 2400

Therefore,

$$10000 + (x - 30000) \times 6/100 - x \times 8/100 = 2400$$

$$10000 + 6x/100 - 1800 - 8x/100 = 2400$$

$$10000 - 1800 - 2400 = 8x/100 - 6x/100$$

$$5800 = 2x/100$$

$$x = 290000$$

Solution:7- Let the total sales be x.

According to the Question,

$$x \times (18/100) + (x - 40000) \times (1/100) = 11000$$

$$(18x)/100 + x/100 - 400 = 11000$$

$$(19x)/100 = 11400$$

$$x = 60000$$

Solution:8- Let the Total marks be x

According to the question,

$$30\% \text{ of } x + 50 = 50\% \text{ of } x - 30$$

$$30x/100 + 50 = 50x/100 - 30$$

$$20x/100 = 80$$

$$x = 400$$

$$\text{Passing marks} = 30\% \text{ of } 400 + 50 = 120 + 50 = 170$$

$$\text{Passing Percentage} = (170/400) \times 100 = 42.5\%$$

Solution:9- Passing marks = 150 + 30 = 180

Let the maximum marks be x

According to the Question

$$40\% \text{ of } x = 180$$

$$x = 180/(40\%) = 450$$

Solution:10-

Total voter in voting list = x

Then Voter's participate in voting

$$\left(x \times \frac{90}{100}\right) \text{ (10\% voter's did not participate in voting)}$$

Then invalid voter

$$\left[\left(x \times \frac{90}{100}\right) - 700\right] \text{ (700 casted voter declared invalid)}$$

Then,

$$\left[\left(x \times \frac{90}{100}\right) - 700\right] \times \frac{40}{100} = 800$$

$$x = \left[\left\{\left(800 \times \frac{100}{40}\right) + 700\right\} \times \frac{100}{90}\right]$$

$$x = (2000 + 700) \times \frac{100}{90}$$

$$x = 3000 \text{ Rs}$$

Solution:11 Let the total number of registered voters = $T = 4000$

Number of voters who did not casted their votes = $x\%$ of Registered voters

So, number of voters who casted their votes = $(100 - x)\%$ of Registered voters

Out of casted votes, 180 were declared invalid.

So, number of valid votes = $(100 - x)\%$ of $T - 180$

Number of votes, winning candidate gets = 42% of T

The number of votes received by losing candidate = $(100 - x)\%$ of $T - 180 - 42\%$ of T
 $= (58 - x)\%$ of $T - 180$

Difference of votes between winner and loser = 42% of $T - [(58 - x)\%$ of $T - 180] = 740$

$(x - 16)\%$ of $T + 180 = 740$

$(x - 16)\%$ of $4000 = 560$

$x - 16 = 14$

$x = 30$

Registered voters who did not casted their votes = $x\%$ of $4000 = 30\%$ of $4000 = 1200$

Hence, option a) is the correct answer.

Solution:12- Let the total voters be 100

No. of votes casted = 85

No. of valid votes = $85 \times (90)/(100) = 76.5$

Winning candidate got 60% of total votes = 60

Looser candidate got = $76.5 - 60 = 16.5$

Difference between winning and looser candidate votes = 3,480

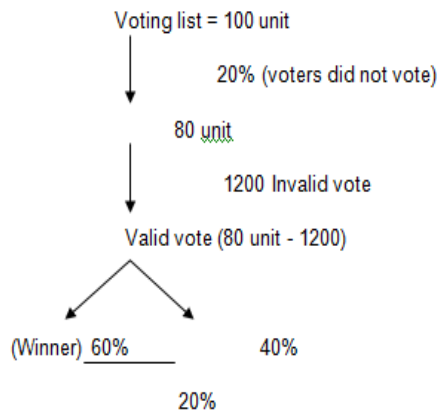
$60 - 16.5 = 43.5$

$43.5 = 3,480$

$100 = 3,480/(43.5) \times 100 = 8,000$

Total number of voters = 8,000

Solution:13-



$$(80 \text{ unit} - 1200) \times (20/100) = 3760$$

$$\{80 \times (20/100)\} \text{ unit} - 1200 \times (20/100) = 3760$$

$$16 \text{ unit} = 3760 + 240$$

$$1 \text{ unit} = 4000/16 = 250$$

$$\text{Voting list} = 100 \text{ unit} = 250 \times 100 = 25000$$

Solution:14- Let the required number = (LCM of denominator 4 and 5) = 20

According to the question,

Wrong answer = $4/5$ times; $20 = 16$

Correct answer = $5/4$ times; $20 = 25$

Required percentage of error = $[(25-16)/25] \text{ times; } 100 = 36\%$

Solution:15 - Ratio of expenditure $\Rightarrow 5: 2 : 3$

We can write these ratio $\Rightarrow 50 \quad 20 \quad 30 \Rightarrow 100$

There is an increment of 10 % $25\% \quad 30\%$

$5 \quad 5 \quad 9 \Rightarrow 19$

Total change will be $\Rightarrow 19 / 100 = 19\%$